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IIIrd SEMESTER

B.E. (IT)

B.E. END SEM. EXAMINATION, Nov.-2019 (ITC07)

Time: 3 Hrs. **DIGITAL CIRCUITS & SYSTEMS** Max. Marks: 40

Note: Attempt all questions. All question carry equal marks. Assume any missing data suitably.

1. Design a 1 DIGIT LCD display that displays the decimal equivalent (5)
of excess 3 codes.
2. Write short notes on the following: (5)
 - a. SRAM and DRAM memory
 - b. ECL and CMOS technology
3. Design NAND gate implementation of the following in its (5)
reduced form.
 $F = A'B'D + C'DB + A'BD + AC'D + ABCD$
Also, represent it into the standard POS.
4. Design a 4 bit universal shift register with following function table. (5)

S1	S0	FUNCTION
0	0	LEFT SHIFT
0	1	RIGHT SHIFT
1	0	PARALLEL LOAD
1	1	NO CHANGE

5. Design a MOD 13 synchronous UP counter with (5)
synchronous reset using D Flip-flops
6. Design a sequence detector for the sequence 0101 and (5)
implement its hardware equivalent by using T flip-flops.
7. Implement the following function using a 8x1 multiplexer. (5)
 $F = \prod(2,3,4,5,6,8,10,12,15)$
8. Explain the functioning of dual slope ADC converter. (5)

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